Matam Padmini

Aspiring AI and Machine Learning Engineer | Artificial Intelligence and Data Science

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padmini-matam.github.io/Portfolio-

Summary

Aspiring AI and Machine Learning Engineer with a Bachelor of Engineering in Artificial Intelligence and Data Science. Hands-on experience in Deep Learning, Large Language Models (LLMs), Retrieval Augmented Generation (RAG), Graph Neural Networks (GNNs), prompt engineering, and Quality Assurance (QA). Worked at Deccan AI and interned at Indian Institute of Technology Hyderabad (IIT Hyderabad). Built Python systems using PyTorch, TensorFlow, LangChain, SQL, and Streamlit, with measurable gains in model accuracy, inference latency, and ranking fairness.

Technical Skills

Languages: Python, SQL, R, C, HTML, CSS, JavaScript
Libraries & Frameworks: scikit-learn, TensorFlow, PyTorch, PyTorch Geometric, NetworkX, Pandas, NumPy, LangChain, ChromaDB, Streamlit, Gradio, Plotly, Sentence Transformers
Tools & Platforms: Jupyter Notebook, VS Code, Git, GitHub, Google Colab, Groq, Hugging Face Spaces
Machine Learning & AI: Machine Learning, Deep Learning, Data Science, CNN, RNN, GCN, GAT, Transfer Learning, RAG, RLHF, LLMs, NLP, Prompt Engineering, Feature Engineering, Model Evaluation, Vector Databases
Core CS: Software-Defined Networking (SDN), Mininet, Ryu SDN Controller, Open vSwitch, sFlow, Computer Networks, Data Structures & Algorithms, DBMS, Object-Oriented Programming (OOP), Data Analytics, Quality Assurance (QA)

Experience

Deccan AI Oct 2025 – Present
AI Operations Associate — AI Product Quality Operations *Hyderabad, India*

- Coordinated end-to-end AI/ML data projects, assigning tasks to freelancers by skill set and workload and tracking productivity, turnaround time, and progress to meet deadlines and quality targets.
- Shared guidelines and updates with freelancers and internal teams, gave specific feedback, and coordinated rework to improve accuracy and consistency.
- Worked on RLHF pipelines, writing complex prompts and evaluating and annotating LLM outputs across text, code, image, and video projects, including VFX stylization.
- Owned prompt design, revision, and quality review for an ML research project alongside a small team, creating adversarial prompts that expose model weaknesses and gold-standard answer sets for each task, with PM guidance.
- Quality-checked prompts and answer sets against source documents, recomputing figures to catch calculation, rounding, and unit errors and flagging guideline deviations before delivery.
- Validated automated verifier checks by inspecting output files (workbooks, memos, PDFs) with Python, separating real model failures from faulty checks and correcting checks that mis-scored valid outputs.

Indian Institute of Technology Hyderabad (IIT Hyderabad) May 2025 – Aug 2025
Research and Development Intern — Air Pollution and Climate Change Modeling *Hyderabad, India*

- Built predictive models for air pollution and climate trends using raw CSV station data from Delhi and Chennai; performed preprocessing, EDA, and visualization in Python.
- Applied Spearman correlation, Mann-Kendall trend tests, and stationarity tests to derive actionable climate insights.
- Delivered findings through structured documentation, directly supporting faculty-led climate modeling research at IIT Hyderabad.

Projects

TalentScout AI – AI Talent Scouting & Engagement Agent *Jun 2026*
Groq Llama 3, Sentence-Transformers, Streamlit GitHub

- Designed an end-to-end agentic recruitment pipeline that automates candidate sourcing, evaluation, AI-to-AI outreach simulation, and shortlist reporting, with CSV, JSON, and PDF export. Live Demo
- Engineered a hybrid ranking engine combining skill overlap, experience, and semantic similarity; integrated a radar chart visualization, GitHub profile importer, and human-in-the-loop recruiter chat.
- Added a Fairness Audit module that flags gender-proxy and location-tier score disparities, improving ranking transparency.

SFlowGCN – Low-Rate Flow Table Overflow Attack Detection in SDN *Sep 2025 – Apr 2026*

PyTorch Geometric, Mininet, Ryu SDN Controller

- Built a real-time graph neural network attack-detection system on an SDN testbed, processing 1.17M+ packet records with a 15-dimensional, 6-feature LFTO-specific node vector.
- Achieved 96.47% accuracy, 99.69% ROC-AUC, and 95.30% TPR, outperforming KNN, GRU, LSTM, and EGST baselines.
- Reduced flow table overflow by 83.4% and Packet-In messages by 63.6% in live deployment, at 0.005ms inference latency.

QA Bot – RAG-Based PDF Question Answering System *Feb 2025 – May 2025*

Groq API (Llama-3.3 70B), HuggingFace Embeddings, LangChain, ChromaDB GitHub

- Built a production-ready RAG pipeline with per-session in-memory vector indexing for complete user isolation; deployed on Hugging Face Spaces with encrypted credential management. Live Demo
- Achieved a 0.4673 F1 score and 0.7130 semantic similarity across a 24-question evaluation spanning three document types, with sub-13s average end-to-end latency.
- Open-sourced the project on GitHub under the MIT License.

Education

Stanley College of Engineering and Technology for Women 2022 – 2026

Bachelor of Engineering in Artificial Intelligence and Data Science — CGPA: 9.36/10 *Hyderabad, India*

Sri Chaitanya Junior College 2020 – 2022

Intermediate (Telangana State Board) — 96.80% *Telangana, India*

Sri Sai Krishna Talent School 2019 – 2020

Secondary School Certificate (SSC) — 100% *Telangana, India*

Certifications

- Google Data Analytics Professional Certificate — Coursera
- AI Fluency: Framework and Foundations — Anthropic
- Object-Oriented Programming using Java and SQL for Data Science — Great Learning
- WE-Hub IDEATHON-2.0 — Idea Validation

Achievements

- Secured 3rd Place at the ACM-W Hackathon for a career recommendation system built with HTML, CSS, and JavaScript. GitHub
- Participated in the Catalyst Hackathon organized by Deccan AI Experts, contributing to innovative AI solutions.
- Built a CNN model with transfer learning, achieving 90% accuracy at the GDSC Hackathon, competing among 50+ teams. GitHub